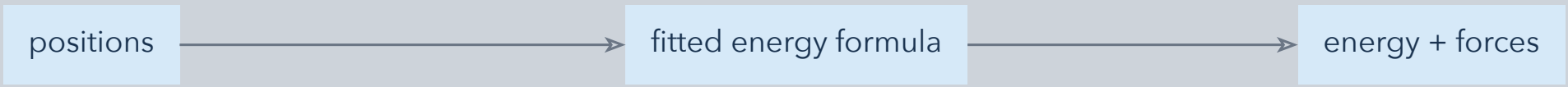
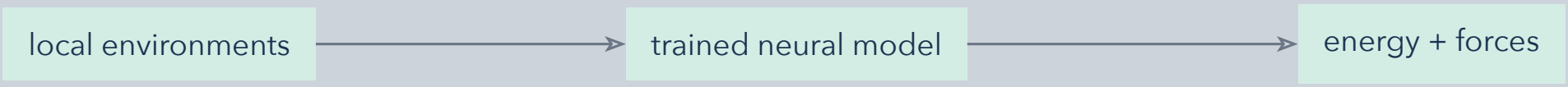
Choose a model for the quantities and configurations your simulation must predict. Speed, accuracy, and transferability depend on the task; they are not universal scores.

1 What is evaluated?

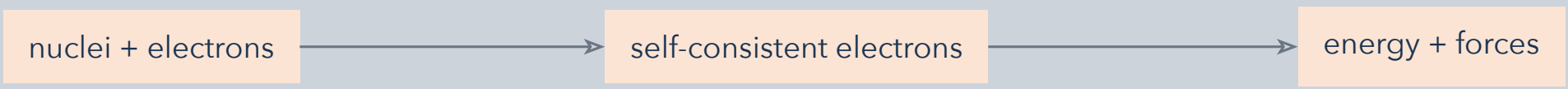
Classical force field



ML potential



Density functional theory



All three can supply energies and forces for atomistic simulation. They differ in how that mapping is built and what electronic physics is represented.

2 What must be validated?

Method	Checks for the intended application
Classical force field	Functional form and parameter coverage; bonding changes, charges, and long-range interactions.
ML potential	Training-domain coverage; errors on representative structures, forces, defects, and trajectories.
DFT	Functional, basis and sampling convergence; relevant charge, spin, and electronic states.

Benchmark the same observable, structures, and accuracy target on the intended hardware. Report throughput and errors with units; test transfer on configurations excluded from fitting.

Ask "good enough for this task?" before "which is best?" Classical and ML models amortize a fitted approximation over many evaluations. DFT solves an approximate electronic problem for each configuration. None guarantees accuracy outside its validated setting.